

Inspiratory Muscle Training Improves Maximal Inspiratory Pressure Without Increasing Performance in Elite Swimmers

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Purpose: To analyze the effect of inspiratory muscle training (IMT) on the maximal inspiratory mouth pressure (MIP) and performance of elite swimmers. **Methods:** Eight participants performed a 3000-m swimming test (T-3000), followed by blood lactate measurements at 1 and 5 minutes postexercise. The testing protocol was carried out before and after 6 weeks of IMT, in which a high-volume IMT group (HV-IMT) ($n = 4$) performed IMT twice a day—in the morning in a seated position and in the afternoon in a concurrent session of IMT and core muscle training. Also, a low-volume IMT group (LV-IMT) ($n = 4$) performed IMT in the morning session only. **Results:** After the intervention, both groups improved their MIP, HV-IMT (132.75 [27.42] to 156.75 [21.88] cmH₂O; $P = .010$; $d = 0.967$) and LV-IMT (149.25 [22.82] to 171.50 [23.74] cmH₂O; $P = .013$; $d = 0.955$), without a significant difference between groups ($P = .855$). Regarding swimming performance, there were no changes between groups in the T-3000 ($P = .472$) or lactate removal rate ($P = .104$). **Conclusion:** IMT increased inspiratory muscle strength in elite swimmers, but there was no association or meaningful impact on swimming performance.

Keywords: respiratory system, respiratory muscle training, swimming, inspiratory muscle strength, lactate removal rate

Inspiratory muscle training (IMT) has been described as an effective means to improve respiratory muscle strength¹ and is widely used in sport in an attempt to enhance athletes' performance.

During exercise, the increased pulmonary ventilation requires intense and prolonged work by the inspiratory muscles, demanding also a rise in the O₂ consumption from these muscles. Some authors have determined the O₂ demand from ventilation during maximal intensity exercise to be ~7% to 10% in healthy, untrained subjects,² and up to 20% in highly trained subjects.³ In this sense, the competition for available blood flow between locomotor muscles and inspiratory muscles during exercise may produce an O₂ deficit over the peripheral muscles, diminishing the aerobic exercise efficiency.⁴ Interestingly, human distribution of cardiac output during heavy exercise may preferentially prioritize the respiratory over the limb muscle vasculature,⁵ in which the respiratory musculature also commands ~15% to 21% of maximal or near-maximal cardiac output.⁴

Comparing different exercise modes, water-based sports are uniquely demanding sports for the inspiratory muscles, compared with land-based sports. Competitive swimming requires the ability to regulate breathing patterns at volumes and flow rates that are much higher than during terrestrial exercise,⁶ including ~2000 to 4000 forced inspirations per training day. Swimming is also characterized by a horizontal body position and thorax compression because of the hydrostatic pressure generated by the water around the chest.⁷ These factors provoke a higher work of breathing in swimming compared with other endurance sports, such as cycling,⁸ which may be linked with the higher lung volume and diffusing capacity shown by swimmers.^{9,10}

Maximal inspiratory mouth pressure (MIP) is defined as the maximum pressure that can be performed by the inspiratory muscles in a forced inspiration, allowing us to evaluate the inspiratory muscles' strength. Some articles have revealed an increase in MIP after IMT,^{6,11} while other studies have not found significant effects.¹² Regarding swimming performance, there are also mixed results, with some studies describing relevant improvements in the sports performance of club-level swimmers⁶ and national-level swimmers,¹³ while others found no influence on swim performance in regional-level swimmers.¹⁴

The latest evidence indicates that the IMT application may not yet be maximized with the current loading protocols, which could constrain the adaptive response in elite athletes.^{15,16} Swimmers may need a higher training load due to their elevated MIP reference values,^{17,18} and higher exposure to forced inspirations during training and competition.¹¹ Then, an optimal protocol should provide a progressive incremental training load,¹⁶ including the application of IMT with sport-specific exercises.¹⁹

Attending to this scientific background, this study aimed to analyze the effect of an IMT protocol on the inspiratory muscle strength, swimming performance, and lactate removal rate of elite long-distance swimmers. This is the first study done with elite long-distance swimmers exposed to a training volume higher than 75 km of swimming per week.

The hypothesis of this study is that IMT would improve the MIP of elite swimmers, accelerating the lactate removal rate, and, therefore, improving swimming aerobic performance.

Methods

Subjects

The participants were 8 elite swimmers—3 females and 5 males (age: 19.6 [2.4] y, height: 176.6 [7.9] cm, weight: 65.5 [5.0] kg, body mass index: 21.0 [0.7], 7-skinfolds sum: 54.1 [10.4] mm) from the Royal Spanish Swimming Federation aged from 17 to

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24 years old. All participants had more than 10 years of swimming training experience, performing more than 35 hours of training per week, and a weekly volume of swim training higher than 75 km (78.57 [5.85] km) during the study. The athletes' swim specialties were 400 m individual medley ($n=4$) and 1500 m ($n=4$) events. The average number of Fédération Internationale de Natation (FINA) points in the participants' best event was 852 FINA points, 3 of them being World Junior finalists, and 3 of them Olympic Swimmers in Tokyo, 2020. All of them were part of the same training group and shared the same type of training (high volume of aerobic training swimming).

Design

Participants were matched by gender and divided into 2 separate groups. The high-volume IMT group (HV-IMT) was composed by 4 swimmers—2 females and 2 males (age: 21.0 [2.6] y, height: 173.9 [8.8] cm, weight: 64.5 [6.3] kg, body mass index: 21.3 [0.8], 7-skinfolds sum: 55.2 [8.8]), while the low-volume IMT group (LV-IMT) was composed of 1 female and 3 males (age: 18.3 [1.3] y, height: 179.4 [7.0] cm, weight: 66.5 [3.9] kg, body mass index: 21.3 [1.1], 7-skinfolds sum: 47.8 [0.4]). The HV-IMT performed twice daily sessions of IMT, one rested session in a seated position, and a concurrent session of core muscle exercises and IMT in a prone position, during the 6 weeks of the study. The LV-IMT only performed the morning session of IMT in a seated position during the study.

To study the effects of IMT on the swimmers' sports physiology and performance, an initial evaluation (PRE) was done before the beginning of the study. The evaluation consisted of an MIP test at rest to evaluate the inspiratory muscles strength, a swimming performance test of 3000-m freestyle (T-3000), and a blood lactate recovery test, immediately following the T-3000, by registering 2 separate samples of lactate, 1 and 5 minutes after the end of the exercise, to evaluate the maximum lactate and the lactate clearance rate. Immediately after the 6 weeks of IMT, there was a second evaluation (POST) following the same procedure, an MIP test, a T-3000, and the subsequent lactate samples at minutes 1 and 5 after the end of the exercise.

Procedures

Maximal Inspiratory Mouth Pressure. The MIP measurement was conducted with a Powerbreathe Kinetic KH1 (IMT Technologies Ltd). A disposable antiviral and antibacterial filter were used in every test.

The participants performed the MIP assessment 1 hour before the T-3000 at the beginning and the end of the 6-week IMT intervention. Prior to the start of the study, the swimmers were asked to participate in a session to become familiar with procedures. These respiratory evaluations followed the international standards from the American Thoracic Society/European Respiratory Society.²⁰ Before the start of the study, the participants were familiarized with the test to achieve a reliable measurement of the MIP. Every participant performed 3 trials of MIP in a seated and resting position, executing the maneuver as quickly and strongly as possible, and maintaining the maximal pressure for at least 1.5 seconds. The best score was collected when a variation lower than 10% among the 3 trials was obtained.

IMT Protocol. A device of resistive load Powerbreathe Plus Competition (IMT Technologies Ltd) was utilized to perform the 6-week IMT intervention. The Powerbreathe Plus Competition

creates a resistance to your in-breath using a calibrated spring loaded "pressure threshold" valve. Every participant was always supervised by a member of the staff during the IMT training program. Both morning and afternoon sessions were conducted before the swimming training as a part of the warm-up. The HV-IMT started the training program at 60% of the MIP, and incremented the load by 10% every 2 weeks to finish the study at an intensity of 80% of the MIP during their twice daily sessions. The LV-IMT maintained a static training load of 60% of the MIP during their single daily session throughout the entire intervention period. The IMT program was included in the Olympic Games planning within the preparatory period, just before an altitude training camp. Due to the fact that all the participants were selected to compete in the National Trials to qualify for the Tokyo 2020 Olympic Games less than 2 months after the study, it was not possible to aim a lower or sham IMT for the LV-IMT.

Therefore, 2 different training doses were conducted by each group:

1. Seated IMT (HV-IMT and LV-IMT)

A morning session of IMT was performed by all study participants (HV-IMT and LV-IMT), 6 days per week, during the 6 weeks of the intervention. The IMT was conducted in a rested and seated position, maintaining a straight posture. The session was composed of 3 sets of 10 forced inspirations as strong and intense as possible, with a resting period of 30 seconds between sets. This proposal of interval training was selected to allow a higher intensity and speed during the executions.

2. Concurrent IMT + core muscle exercises (HV-IMT only)

An afternoon concurrent session of IMT + core muscle exercises was performed by the HV-IMT, 6 days per week, during the 6 weeks of the intervention. This training session consisted of 3 sets of 4 exercises of core muscle training, where each of them had a duration of 15 breathing duration, measured by the Powerbreathe Plus Competition, and 15 seconds of rest between exercises. The core exercises were varied during the IMT intervention to maintain an adequate training stimulus. They were directed by the fitness coach of the team and performed in a prone and lateral position to maintain the sport-specific horizontal position of swimming, as Karsten et al²¹ suggested. The program included static and dynamic core exercises, involving shoulder, hip, and trunk movements. The LV-IMT performed the same core muscle exercises as part of the training program, without using the Powerbreathe Plus Competition to perform IMT. This concurrent IMT increased the training volume of the HV-IMT group in 135 breathings following the recommendation made by Shei,¹⁹ which considers that swimmers need a supplementary respiratory training load to generate significant changes in lung function.

Swimming Performance in T-3000. A maximal intensity T-3000 was performed at the beginning and end of the 6-week IMT intervention. This test was selected to evaluate the long-distance swimmers' aerobic fitness and was conducted in a 25-m swimming pool. Before the swimming test, a standardized warm-up was carried out according to the head coach's instructions. The T-3000 was performed in an indoor swimming pool where the swimmers trained with a water temperature of 27 °C. During the trial, swimmers used competition swimsuits with the objective of simulating a competitive environment and minimizing interindividual differences due to the use of different swimsuits, weight, size, and friction. The final time was registered by the coach with a

handheld stopwatch Interval 2000 Split/Rate Watch from Nielsen-Kellerman Inc.

Lactate Test. Blood lactate samples were obtained 1 and 5 minutes after the end of the swimming test, by an experimented person at the High Performance Center of Sant Cugat. A small puncture was performed in the ear lobe with a sterilized lancet. Then, a portable analyzer Lactate Pro 2 (ARKRAY) was used to collect 8 μL of capillary blood using a test strip. The result was presented 15 seconds after the procedure in millimoles per liter ($\text{mmol}\cdot\text{L}^{-1}$).

Statistical Analysis

Data are reported as mean (SD). Relative changes are also presented in percentage (%) to describe the magnitude of variation in some physiological and performance parameters. First, a Shapiro-Wilk test was used to establish the normal distribution of the sample, and Levene test was used to assess for homogeneity of variance between groups. Differences in MIP, T-3000, and lactate removal rate between preintervention and postintervention conditions were analyzed using a 2-way repeated-measures analysis of variance with 2 independent factors: group intersubjects (HV-IMT vs LV-IMT) and time intrasubjects (PRE vs POST). In the case of detecting statistical effects ($P < .05$), Bonferroni comparisons were performed. Effect size (Cohen d) was calculated to estimate the magnitude of the difference between group means, with $d = 0.1, 0.3, 0.5, 0.7$, and 0.9 reflecting small, medium, large, very large, and extremely large effect sizes, respectively.²² The software package used for the statistical analysis was SPSS (version 26, IBM SPSS Statistics).

Ethical Considerations

The study protocol and procedures were approved by the Ethical Committee of the Universidad Isabel I de Castilla (482–21), and the study followed the principles of the Declaration of Helsinki for human experimentation. All the participants were informed of the purpose, protocol, and procedures before informed consent was obtained from them or their representatives.

Results

Maximal Inspiratory Mouth Pressure

The baseline MIP of the entire sample of elite long-distance swimmers were higher than those predicted by age and height ($141.00 [24.96] \text{ cmH}_2\text{O}$; 122%), using the reference data reviewed by Sclauser Pessoa et al.²³ After the intervention, there was a significant increase in the MIP of the entire group of

swimmers, increasing the percentage of predicted values by age and height up to 142% ($164.13 [22.56] \text{ cmH}_2\text{O}$). The comparisons between conditions (group $\times$ time) are shown in Table 1. Both groups increased the MIP from preintervention to postintervention, respectively HV-IMT ($\Delta 15.3\%$; $P = .010$; $d = 0.967$) and LV-IMT ($\Delta 13.0\%$; $P = .013$; $d = 0.955$). However, there were no statistical differences between groups from preintervention to postintervention, respectively ($P = .855$; statistical power [sp] = 0.051). Figure 1 shows the intraindividual changes in the MIP after the intervention, which might be interesting when contemplating the intraindividual response to IMT in every swimmer.

Swimming Performance Test

Changes in the T-3000 after the 6-week intervention are shown in Table 2. There were no changes in T-3000 after the IMT in any of the studied groups, HV-IMT ($\Delta 0.89\%$; $P = .238$; $d = 0.214$) and LV-IMT ($\Delta 2.19\%$; $P = .224$; $d = 0.650$). In the comparison between groups, there were no significant differences in HV-IMT and LV-IMT, respectively ($\Delta 17.00 [23.11]$ and $\Delta 43.75 [57.30]$ s; $P = .236$; sp = 0.183).

Postexercise Lactate Recovery

Changes in the lactate removal rate after the 6-week IMT are shown in Table 3. After the intervention, there were no significant changes in the lactate removal rate in any of the studied groups of swimmers, HV-IMT ($\Delta 15.40\%$; $P = .247$; $d = 0.733$) and LV-IMT ($\Delta -4.14\%$; $P = .600$; $d = 0.637$). In the comparison between groups, there were no significant differences in HV-IMT and LV-IMT, respectively ($P = .338$; sp = 0.129).

Discussion

Maximal Inspiratory Mouth Pressure

Elite long-distance swimmers presented a high basal MIP compared to the reference values for their age and height (122%). Despite this, and their usual weekly high volume of training ($>75 \text{ km}$), a 6-week intervention of IMT improved the MIP ($\Delta 14.1\%$) in the entire swimming training group. This result suggests that a minimum dose of IMT improves MIP, even in athletes with highly developed inspiratory muscle strength.

Until now, only one study had studied elite swimmers⁹ with related, but much larger effects (67% increased MIP after 12 weeks of IMT). However, such an increment could be also linked to the increase in physical fitness of the swimmers during the period of intervention since the study was conducted at the beginning of the

Table 1 Changes in MIP Before and After the 6-Week IMT in Both Groups Studied, HV-IMT and LV-IMT

	HV-IMT (n = 4)		LV-IMT (n = 4)		P (sp)
	Pre	Post	Pre	Post	
MIP (cmH_2O)	132.75 (27.42)	156.75 (21.88)*	149.25 (22.82)	171.50 (23.74)*	.855 (0.051)
Δ MIP (cmH_2O)		24.00 (16.43)		22.25 (8.26)	
Δ MIP, %		15.32 (16.43)		12.98 (8.26)	

Abbreviations: HV-IMT, high-volume IMT group; IMT, inspiratory muscle training; LV-IMT, low-volume IMT group; MIP, maximal inspiratory pressure; Δ MIP, variation of the MIP from preintervention to postintervention; P value, the statistical probability of a significant effect in the comparison "group $\times$ condition"; sp, statistical power. Note: Results are presented as mean (SD).

*Significant differences (P value $< .05$) between precondition and postcondition intragroup.

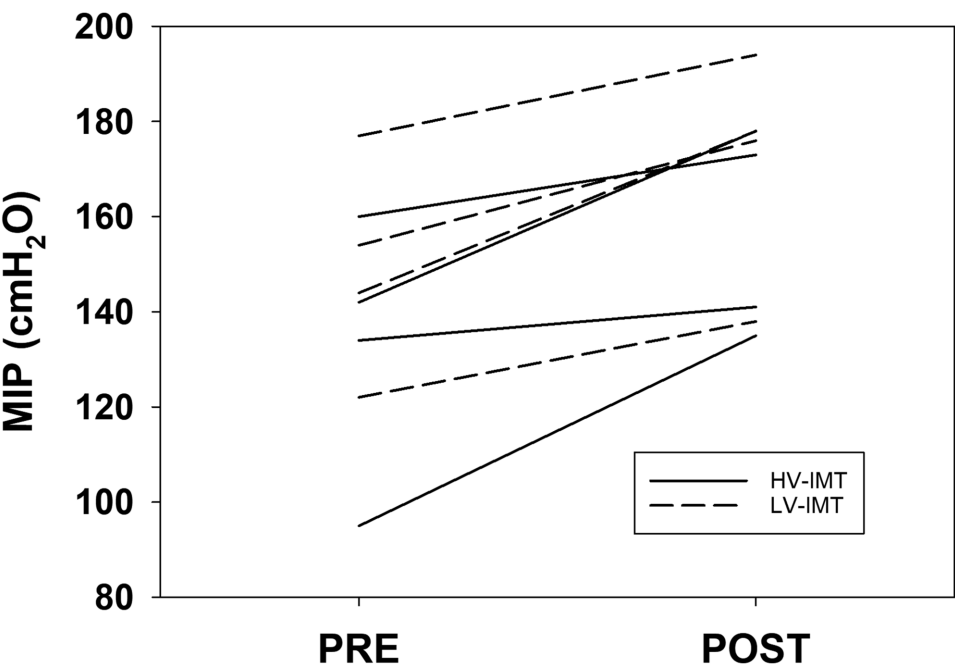


Figure 1 — Intraindividual changes of MIP before and after the 6-week IMT intervention in both groups, HV-IMT and LV-IMT. HV-IMT indicates high-volume IMT group; IMT, inspiratory muscle training; LV-IMT, low-volume IMT group; MIP, maximal inspiratory pressure.

Table 2 Changes in Swimming Performance in the T-3000 Before and After the 6-Week IMT in Both Groups Studied, HV-IMT and LV-IMT

	HV-IMT (n = 4)		LV-IMT (n = 4)		P (sp)
	Pre	Post	Pre	Post	
T-3000, min:s	32:34.50 (01:09.26)	32:17.50 (01:28.49)	33:26.00 (00:49.41)	32:42.25 (01:21.34)	.236 (0.186)
Δ T-3000, s	17.00 (23.11)		43.75 (57.30)		
Δ T-3000, %	−0.89 (1.20)		−2.19 (2.85)		

Abbreviations: HV-IMT, high-volume IMT group; IMT, inspiratory muscle training; LV-IMT, low-volume IMT group; *P* value, the statistical probability of a significant effect in the comparison “group × condition”; sp, statistical power; T-3000, swim performance in a 3000-m test; Δ T-3000, variation of the T-3000 from preintervention to postintervention. Note: Results are presented as mean (SD).

Table 3 Changes in Lactate Removal Rate Before and After the 6-Week IMT in Both Groups Studied, HV-IMT and LV-IMT

	HV-IMT (n = 4)		LV-IMT (n = 4)		P (sp)
	Pre	Post	Pre	Post	
Lac 1, mmol·L ^{−1}	2.98 (0.49)	3.75 (1.51)	2.65 (0.33)	3.72 (0.56)	.338 (0.126)
Lac 5, mmol·L ^{−1}	2.33 (0.88)	2.50 (1.67)	2.03 (0.21)	3.00 (0.32)	
Lactate removal rate, %	23.75 (19.65)	39.16 (21.96)	23.04 (8.72)	18.90 (6.84)	
Δ Lactate removal rate, %	15.40 (21.49)		−4.14 (14.20)		

Abbreviations: HV-IMT, high-volume IMT group; IMT, inspiratory muscle training; Lac 1, lactate production 1 minute after the end of the 3000-m swim test; Lac 5, lactate production 5 minutes after the end of the 3000-m swim test; Lactate removal rate, blood lactate decline from Lac 1 to Lac 5 for every condition studied; Δ Lactate removal rate, comparison of the lactate removal rate from preintervention to postintervention; LV-IMT, low-volume IMT group; *P* value, the statistical probability of a significant effect in the comparison “group × condition”; sp, statistical power. Note: Results are presented as mean (SD).

season after the detraining phase. In that case, all the groups (IMT, sham-IMT, and controls) showed similar increases in the MIP.

Interestingly, the swimmers from the study of Mickleborough et al⁹ had higher basal MIP compared with the data from our study (180.2 [27.9] vs 141.00 [24.96] cmH₂O), despite the lower weekly volume of swimming (40–60 vs 78 km). This evidence suggests that the inspiratory muscle strength might be more influenced by anthropometric parameters, athletes’ swim speciality, or other factors than their swimming training volume per se.¹⁷

These results also agree with the findings by Kilding et al,⁶ who found an increase in inspiratory muscle strength after a 6-week IMT at 50% of the MIP in regional-level swimmers. Shei et al¹² also observed an increase in MIP and mean maximal voluntary ventilation after a 12-week intervention of IMT with subelite swimmers. Also, Lomax et al¹¹ and Kapus¹³ have shown increases in the inspiratory muscles' strength after 6 weeks of IMT, 2 sessions per day at 50% of the MIP, with national-, international-, and regional-level swimmers, respectively.

On the other hand, Shei et al¹² did not find an increase in inspiratory muscle strength after 4 weeks of IMT. The selected protocol in their study was 3 days per week, at 80% of the MIP with progressively increased work–rest ratios until task failure. This study proposed a different protocol than commonly used in the literature, with lower training volume and high intensity. These results might suggest that a minimum volume dose is necessary to increase MIP.^{15,19}

From our data, it can be interpreted that a daily dose of IMT at 60% of the MIP is enough to generate adaptations on the inspiratory muscle strength in elite swimmers with a high volume of training. In this sense, a higher IMT dose seems to not provide an additional effect, although further studies should explore the dose–effect relationship in swimmers with a lower regimen of training volume and/or higher sample sizes, since the variability of the interindividual response is reasonable high (Figure 1).

Swimming Performance Test

From a statistical point of view, swimming performance was not significantly improved in the T-3000 after the IMT. In this sense, it must be acknowledged that *sp* was low (0.186), perhaps leading to a type II error. Also, the entire swimming group decreased by 30 seconds the time in the T-3000 (–1.5%), a difference that has a relevant impact in a sample of elite international-level swimmers. In comparison with other studies, Kilding et al⁶ found that IMT provokes a significant increase in the swim performance of events shorter than 400 m, specifically 100 and 200 m in club-level swimmers (–1.7% and –1.5%, respectively). Lomax et al¹¹ reported that swimming performance in 100- and 200-m swimmers was only improved when weekly swimming training volume was no larger than 31 km (national level), compared with the group that trained more than 41 km per week (international level). Kapus¹³ also found a small significant improvement in swimming speeds in the 50-m butterfly and 100-m front crawl, but not in the 50-m front crawl and 50-m breaststroke. However, Cunha et al¹⁴ did not show any significant change in swimming performance after a 12-week IMT intervention in regional-level swimmers (average 459 FINA points).

Until now, this is the only study that has evaluated the impact of IMT on the swim performance of elite long-distance swimmers and we found no significant change in swimming performance after IMT. However, the *sp* of the analysis was low (*sp* = 0.186) due to the limited sample size of elite swimmers participating in the study. Our sample has an elite international level, including 3 Olympic semifinalists and 3 World Junior Champions finalists, who have a weekly swimming training routine far more intense than in any other study described (78.57 [5.85] km). Therefore, the high level of swimming performance and high loads of aerobic training could mask any the effect of IMT on swimming performance parameters since the magnitude of improvement of these physiological parameter is limited at this level of performance.

Last of all, a systematic review and meta-analysis conducted by Karsten et al¹⁵ also observed the lack of meaningful impact of IMT on

the sports performance of aquatic athletes, regardless of the increases in MIP. The reason for this is unclear, but could, perhaps, be related to an already elevated basal MIP in swimmers, or, the limited influence of inspiratory muscle strength on aquatic sports performance.

Lactate Removal Rate

In our study, there were no significant changes in the postexercise lactate removal rate after the 6-week IMT. Our initial hypothesis was that an increase in inspiratory muscle strength would accelerate the lactate removal rate by increasing the density of the transport protein of lactate through the lactate shuttle in the inspiratory muscles,²⁴ the monocarboxylate transporter type 1.²⁵ In our data, a nonsignificant tendency could be appreciated comparing HV-IMT and LV-IMT, respectively ($\Delta 15.40$ [21.49%] and $\Delta -4.14$ [14.20%]; $P = .338$). Despite the promising results, there was large interindividual variability in the HV-IMT, reducing the physiological concordance of these changes.

Some studies in the literature have supported this initial hypothesis. Chiappa et al²⁶ found that a slight resistive load in the inspiratory muscles during postexercise recovery provokes a faster lactate removal in healthy subjects, concluding that the inspiratory muscles may be considered a net lactate consumer. Rozek-Piechura et al²⁷ found a significant reduction in maximum lactate production after an incremental test in runners after 8 weeks of IMT. However, Kilding et al⁶ observed that the performance-enhancing influence of IMT is not due to an improvement in lactate metabolism since there was no substantial change in swimming speed at the lactate threshold.

Limitations

A potential reason to explain the lack of significant results in our study could be the high fitness baseline level of these swimmers, who are exposed to high loads of aerobic training on a daily basis, which is directly associated with an efficient lactate removal rate. The low number of subjects studied is another limitation, insurmountable, and common to many studies with elite athletes. This study does not provide an evaluation of inspiratory muscle fatigue. This would have been useful to ascertain if the improvements in strength are translated into less fatigue given that swim time did not significantly change. Also, it is reasonable to think that core muscle training may increase the strength of the expiratory muscles, hence an analysis of maximal expiratory pressure might have been interesting.

Practical Applications

The increases in the inspiratory muscle strength have been relevant enough to consider investigating the effects of IMT in the sport of swimming, although the use of IMT did not improve swimming performance. Nevertheless, at this level of human performance, an increase of 14% of any physiological variable should be considered as meaningful. Such a change could facilitate the breathing pattern, promoting the hydrostatic body position while swimming, and delaying the inspiratory muscles fatigue.

Conclusions

Inspiratory muscle training increased inspiratory muscle strength in elite swimmers, but a higher training volume and intensity in IMT does not provide an additional effect on inspiratory muscle strength. In addition, there was a meaningful, although not statistically significant,

improvement in swimming performance in the T-3000 after the IMT in a sample of elite long-distance swimmers (-1.5% ; $d = 0.650$).

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